

- plan:
- finish pointwise convergence proof when $f'(x_0)$ exists
 - Riemann-Lebesgue lemma
 - pointwise convergence at a jump discontinuity

recap: we are partway done proving:

Theorem: (pointwise convergence, version 1)

If $f(x)$ is continuous and 2π -periodic, then for each x_0 where f is differentiable, $S_N(x_0) \rightarrow f(x_0)$ as $N \rightarrow \infty$.

steps so far:

$$S_N(x_0) = P_N f(x_0) = \sum_{k=-N}^N c_k e^{ikx_0}, \quad c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx$$

$$= \int_{-\pi}^{\pi} f(x) D_N(x_0 - x) dx$$

Dirichlet kernel: $D_N(x) = \frac{1}{2\pi} \sum_{k=-N}^N e^{ikx}$

$$\int_{-\pi}^{\pi} D_N(x) dx = 1$$

$$\text{for } x \in [-\pi, \pi], \quad 2\pi D_N(x) = \begin{cases} \frac{\sin((N+\frac{1}{2})x)}{\sin(\frac{x}{2})}, & x \neq 0 \\ 2N+1, & x = 0 \end{cases}$$

$D_N(x)$ is 2π -periodic, even, and continuous (even at $x=0$)

$$S_N(x_0) = \int_{-\pi}^{\pi} f(x_0 + u) D_N(u) du \quad (\text{change of variable})$$

Since $\int_{-\pi}^{\pi} D_N(x) dx = 1$, we have

$$\begin{aligned} S_N(x_0) - f(x_0) &= \int_{-\pi}^{\pi} \left[f(x_0 + u) - f(x_0) \right] \underbrace{D_N(u)}_{\frac{1}{2\pi} \frac{\sin((N+\frac{1}{2})u)}{\sin(u/2)}} du \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} g(u) \sin\left(N + \frac{1}{2}\right)u \, du \end{aligned}$$

where

$$g(u) = \begin{cases} \frac{f(x_0 + u) - f(x_0)}{u} \cdot \frac{u}{\sin(u/2)} & u \neq 0, |u| \leq \pi \\ g(0) & u = 0 \end{cases}$$

(transfer bad denominator to other factor)

$f(x)$ continuous $\Rightarrow g(u)$ is continuous (except possibly at $u=0$)

$$\lim_{u \rightarrow 0} g(u) = f'(x_0) \underbrace{\left(\lim_{u \rightarrow 0} \frac{1}{\frac{1}{2} \cos \frac{u}{2}} \right)}_{\text{L'Hospital's rule}} = 2f'(x_0)$$

so setting $g(0) = 2f'(x_0)$ makes $g(u)$ continuous on $[-\pi, \pi]$.

In general, $g(-\pi) = -g(\pi) \neq g(\pi)$ (not periodic)

The Riemann-Lebesgue lemma now implies that

$$\lim_{N \rightarrow \infty} [S_N(x_0) - f(x_0)] = 0, \quad \text{i.e.} \quad \lim_{N \rightarrow \infty} S_N(x_0) = f(x_0)$$

R-L lemma: If $f \in L^1(a, b)$, then

$$\lim_{k \rightarrow \infty} \int_a^b f(x) e^{-ikx} dx = 0$$

can extend to $\mathbb{R}$
via $f(x) = 0$

$$\lim_{k \rightarrow \infty} \int_a^b f(x) \cos kx dx = 0$$

for $x < a$ or $x > b$
(the lemma is really about $f \in L^1(\mathbb{R})$)

$$\lim_{k \rightarrow \infty} \int_a^b f(x) \sin kx dx = 0$$

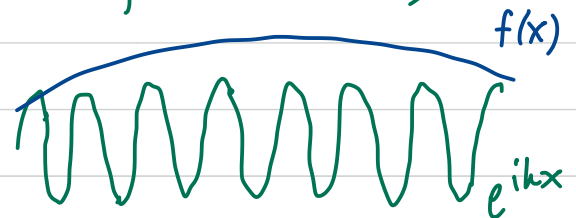
where k is not restricted to be an integer.

our case: version 3 with $k = N + \frac{1}{2}$, $a = -\pi$, $b = \pi$, $f = \frac{g}{2\pi}$

Idea behind R-L lemma:

for large k , e^{ikx} oscillates fast in comparison to $f(x)$,

so there is a lot of cancellation:



proof of R-L lemma ^(version 3) when f is C^1 (continuously differentiable) and (a, b) is finite:

$$\int_a^b f(x) \sin kx dx =$$

$$u = f, \quad dv = \sin kx dx$$

$$= -\frac{\cos kx}{k} f(x) \Big|_a^b + \int_a^b \frac{\cos kx}{k} f'(x) dx$$

$$v = -\frac{1}{k} \cos kx$$

$$= -\frac{\cos kb}{k} f(b) + \frac{\cos ka}{k} f(a) + \frac{1}{k} \int_a^b \cos kx f'(x) dx$$

$$\left| \int_a^b f(x) \sin kx \, dx \right| \leq \frac{1}{k} \left(|f(b)| + |f(a)| + \int_a^b |f'(x)| \, dx \right) \\ \rightarrow 0 \text{ as } k \rightarrow \infty.$$

The case when f is piecewise C^1 can be broken up into intervals:

$$\int_a^b f(x) \sin(kx) \, dx = \sum_{j=1}^J \int_{x_{j-1}}^{x_j} f(x) \sin kx \, dx \rightarrow 0 \text{ as } k \rightarrow \infty \\ (\text{by the } C^1 \text{ proof above})$$

piecewise C^1 means $f(x)$ and $f'(x)$ are continuous on the open intervals (x_0, x_1) , (x_1, x_2) , $\dots$, (x_{J-1}, x_J) where $a = x_0 < x_1 < x_2 < \dots < x_J = b$, and the following one-sided limits exist:

$$f(x_j^+) = \lim_{x \rightarrow x_j^+} f(x) \quad 0 \leq j \leq J-1$$

$$f'(x_j^+) = \lim_{x \rightarrow x_j^+} f'(x) \quad 0 \leq j \leq J-1$$

$$f(x_j^-) = \lim_{x \rightarrow x_j^-} f(x) \quad 1 \leq j \leq J$$

$$f'(x_j^-) = \lim_{x \rightarrow x_j^-} f'(x) \quad 1 \leq j \leq J$$

We do not assume $f(x_j^-) = f(x_j^+)$ or $f'(x_j^-) = f'(x_j^+)$.

The case of $\cos kx$ is similar. (see book)

The case of e^{ikx} follows from these via $e^{ikx} = \cos kx + i \sin kx$

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The full Riemann-Lebesgue lemma asserts that if $f \in L^1(\mathbb{R})$,

then $\hat{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$ belongs to $C_0(\mathbb{R})$, the space of continuous functions on $\mathbb{R}$ for which

$$\lim_{k \rightarrow \infty} \hat{f}(k) = 0 \quad \text{and} \quad \lim_{k \rightarrow -\infty} \hat{f}(k) = 0.$$

continuity: let $k \in \mathbb{R}$ and $\{k_j\}$ be a sequence $k_j \rightarrow k$.

$$\begin{aligned} \text{Then } \hat{f}(k_j) &= \int_{-\infty}^{\infty} f(x) e^{-ik_j x} dx \\ &\rightarrow \int_{-\infty}^{\infty} f(x) e^{-ikx} dx = \hat{f}(k) \quad \text{as } j \rightarrow \infty \end{aligned}$$

where we used the dominated convergence theorem:

$$f(x) e^{-ik_j x} \rightarrow f(x) e^{-ikx} \quad \text{pointwise in } x \text{ as } j \rightarrow \infty$$

$$\text{and } |f(x) e^{-ik_j x}| \leq \underbrace{|f(x)|}_{\text{dominator}} \text{ which is integrable}$$

since $\hat{f}(k_j) \rightarrow \hat{f}(k)$ for every sequence $k_j \rightarrow k$,

$\hat{f}(k)$ is continuous at every $k \in \mathbb{R}$.

zero limit as $k \rightarrow \infty$ and $k \rightarrow -\infty$: here we use a theorem from real analysis that there is a sequence of compactly supported C^1 functions $f_n(x)$ that converges to $f(x)$ in L^1 .

Each $\hat{f}_n(k)$ is continuous (by the above argument) and converges to zero as $k \rightarrow \infty$ and as $k \rightarrow -\infty$ (by our proof for the C^1 case on a finite interval (a, b)).

So each $\hat{f}_n \in C_0(\mathbb{R})$. Next we claim that $\hat{f}_n \rightarrow \hat{f}$ in the uniform norm:

$$\begin{aligned} |\hat{f}_n(k) - \hat{f}(k)| &= \left| \int_{-\infty}^{\infty} [f_n(x) - f(x)] e^{-ikx} dx \right| \\ &\leq \int_{-\infty}^{\infty} |f_n(x) - f(x)| \underbrace{|e^{-ikx}|}_1 dx = \|f_n - f\|_1 \end{aligned}$$

$$\text{so } \|\hat{f}_n - \hat{f}\|_{\infty} \leq \|f_n - f\|_1 \rightarrow 0 \text{ as } n \rightarrow \infty.$$

One can show that $C_0(\mathbb{R})$ is closed in the uniform norm, so $\hat{f}(k) \in C_0(\mathbb{R})$, which is what Riemann-Lebesgue asserts.

For more details, e.g. the proof that $C_0(\mathbb{R})$ is closed in the uniform norm, see Folland, Real Analysis.

Convergence of Fourier series at a point of discontinuity

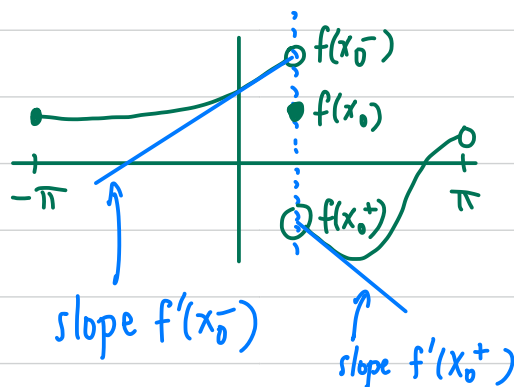
def: $f: [a, b] \rightarrow \mathbb{C}$ is piecewise continuous if it has only finitely many discontinuities $x_1, \dots, x_n$, all of which are jumps. $\left(\begin{matrix} f(x_j^\pm) \\ \text{exist} \end{matrix} \right)$

definitions:

$$f(x_0^\pm) = \lim_{h \rightarrow 0^\pm} f(x_0 + h)$$

$$f'(x_0^\pm) = \lim_{h \rightarrow 0^\pm} \frac{f(x_0 + h) - f(x_0^\pm)}{h}$$

↑
(slight notation conflict with C^1 definition)



note: {

- We do not assume $f(x)$ exists for $x \neq x_0$
- We did not define $f'(x_0^\pm)$ as $\lim_{h \rightarrow 0^\pm} f'(x_0 + h)$
- but if this limit exists, it does equal $f'(x_0^\pm)$ from our definition. (proof: use mean value theorem)

Theorem: Suppose $f(x)$ is piecewise continuous on $[-\pi, \pi]$ and $\underline{f(x_0^\pm)}, f'(x_0^\pm)$ exist. Then

(redundant) →

$$\underline{S_N(x_0)} \rightarrow \frac{1}{2}(f(x_0^+) + f(x_0^-)) \text{ as } N \rightarrow \infty.$$

$$\sum_{k=-N}^N c_k e^{ikx_0}, \quad c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx$$

proof of theorem:

$$\text{still have } S_N(x_0) = \int_{-\pi}^{\pi} f(x_0+u) D_N(u) du, \quad D_N(u) = \frac{1}{2\pi} \frac{\sin((N+\frac{1}{2})u)}{\sin(\frac{u}{2})}$$

The problem now is that $\lim_{u \rightarrow 0} \frac{f(x_0+u) - f(x_0)}{u}$ may not exist.

Averaging technique: since $D_N(u)$ is even, we also have

$$S_N(x_0) = \int_{-\pi}^{\pi} f(x_0-v) D_N(v) dv \quad (v = -u)$$

$$\Rightarrow S_N(x_0) = \int_{-\pi}^{\pi} \frac{f(x_0+u) + f(x_0-u)}{2} D_N(u) du$$

$$\text{as } u \rightarrow 0^+, \quad \begin{aligned} f(x_0+u) &\rightarrow f(x_0^+) \\ f(x_0-u) &\rightarrow f(x_0^-) \end{aligned}$$

$$\text{as } u \rightarrow 0^-, \quad \begin{aligned} f(x_0+u) &\rightarrow f(x_0^-) \\ f(x_0-u) &\rightarrow f(x_0^+) \end{aligned}$$

they switch
but the average
is the same

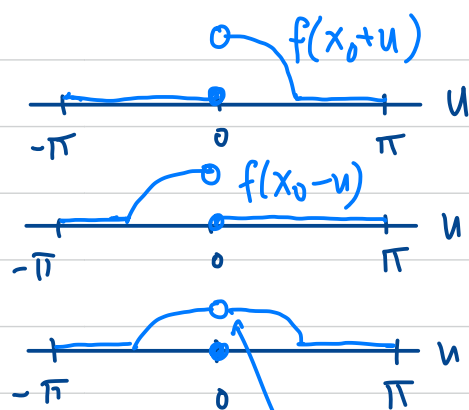
so $\frac{f(x_0+u) + f(x_0-u)}{2}$ becomes

continuous at $u=0$ if we redefine

it to be $\frac{1}{2}(f(x_0^+) + f(x_0^-))$ at $u=0$

→ now the limiting δ -function $D_N(u)$

as $N \rightarrow \infty$ knows what value to select!



$\frac{f(x_0+u) + f(x_0-u)}{2}$
has a limit
as $u \rightarrow 0$

To make this rigorous, we use $\int_{-\pi}^{\pi} D_N(u) du = 1$ to obtain

$$S_N(x_0) - \frac{f(x_0^+) + f(x_0^-)}{2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(u) \sin\left(\left(N + \frac{1}{2}\right)u\right) du$$

where

$$g(u) = \begin{cases} \frac{f(x_0+u) + f(x_0-u) - f(x_0^+) - f(x_0^-)}{2u} \cdot \frac{u}{\sin(u/2)} & u \neq 0 \\ 0 & u = 0 \end{cases}$$

g has a jump discontinuity at $u=0$:

$$\begin{aligned} g(0^+) &= \lim_{u \rightarrow 0^+} \frac{1}{2} \left[\frac{f(x_0+u) - f(x_0^+)}{u} - \frac{f(x_0-u) - f(x_0^-)}{-u} \right] \cdot \frac{1}{1/2} \\ &= f'(x_0^+) - f'(x_0^-) \end{aligned}$$

$$\begin{aligned} g(0^-) &= \lim_{u \rightarrow 0^-} \frac{1}{2} \left[\frac{f(x_0+u) - f(x_0^-)}{u} - \frac{f(x_0-u) - f(x_0^+)}{-u} \right] \frac{1}{1/2} \\ &= f'(x_0^-) - f'(x_0^+) = -g(0^+) \end{aligned}$$

But $g(u)$ is still piecewise continuous, and hence in $L^1(-\pi, \pi)$

→ The Riemann-Lebesgue lemma still applies, and

$$\lim_{N \rightarrow \infty} \left(S_N(x_0) - \frac{f(x_0^+) + f(x_0^-)}{2} \right) = \lim_{N \rightarrow \infty} \frac{1}{2\pi} \int_{-\pi}^{\pi} g(u) \sin\left(\left(N + \frac{1}{2}\right)u\right) du = 0$$

↑
the oscillations lead to more and more cancellation as $N \rightarrow \infty$.

Note that $S_N(x) = \sum_{k=-N}^N C_k e^{ikhx}$ is 2π -periodic,

so outside of $(-\pi, \pi]$, $S_N(x)$ converges to the periodic extension of $f(x)$ from $(-\pi, \pi]$.

$$f_p(x) = f(x - 2\pi m) \quad \leftarrow \begin{array}{l} \text{choose } m \text{ so that} \\ -\pi < x - 2\pi m \leq \pi \end{array}$$

$$\text{Let } F(x) = \lim_{N \rightarrow \infty} S_N(x). \quad \left(\begin{array}{l} \text{It's a pointwise limit.} \\ \text{Each } x \text{ involves} \\ \text{a separate limit.} \end{array} \right)$$

Suppose f is piecewise C^1 on $[-\pi, \pi]$. Then

$$F(\pi) = \frac{1}{2} \left(\overbrace{f(-\pi^+)}^{f_p(\pi^+)} + \overbrace{f(\pi^-)}^{f_p(\pi^-)} \right) \quad \leftarrow f_p(x) \text{ may have a jump at } x = \pi$$

$$F(x_0) = f(x_0) \quad \text{if } x_0 \in (-\pi, \pi), f \text{ cont. at } x_0$$

$$F(x_0) = \frac{1}{2} (f(x_0^+) + f(x_0^-)) \quad \text{if } x_0 \in (-\pi, \pi) \text{ in general}$$

$$F(x) = F(x - 2\pi m) \quad x \in ((2m-1)\pi, (2m+1)\pi], \quad k \in \mathbb{Z}$$

$\leftarrow (x - 2\pi m \in (-\pi, \pi])$

$\leftarrow \text{e.g. } F(-\pi) = F(\pi), m = -1$